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A New Ultraslim LWD Solution Unlocks Formation Evaluation and Geosteering for Underbalance Coiled Tubing Drilling

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Abstract

Underbalance coiled tubing drilling provides an effective way to produce hydrocarbon by minimizing reservoir damage while drilling. The ultra-slim small borehole size (3.5”) however precludes the use of standard logging-while-drilling (LWD) technology which has been developed for "slim" boreholes starting from a 5.5” bit size. There was a concern that the lack of LWD measurements in underbalance coiled-tubing drilling was leading to drilling wells blindly. Sparse mud logging data and gamma ray were the only measurements available to make decisions about well placement to maximize the well net-to-gross ratio. To fill this gap, a unique workflow was designed to integrate the latest LWD technology with mud logging and offset wells data. The workflow provides formation evaluation answers and real-time geosteering information to place the borehole in the sweet reservoir spot.

The ultra-slim LWD resistivity was recently introduced in the field for logging while drilling in wells with underbalance coiled tubing operations. It was designed to withstand the very harsh drilling environment, including high shocks and vibrations and high temperature. The tool is a compensated electro-magnetic propagation measurement, providing phase shift and attenuation at two different frequencies. The four resistivity measurements are combined with the gamma ray measurement into a model-compare-update workflow to take real-time decisions while drilling on the wellbore trajectory positioning. A near-real-time machine-learning workflow provides formation porosity and mineralogy using the medium and deep resistivity and gamma ray from offset wells and advanced mud logging. This workflow integrates x-ray fluorescence and diffraction measured on cuttings, although the data is very sparse owing to the challenging cuttings collection conditions. It is calibrated to dynamic well data.

The LWD resistivity measurement was validated by comparison with offset wells resistivity logged in overbalanced condition with conventional wireline and LWD measurement in a test well and by comparison with a wireline run conveyed on coiled tubing in one well. The workflow was applied on multiple wells in clastic and carbonate environments. Based on the observed production data, the geosteering workflow was deemed successful. The ultra-slim LWD resistivity tool for coiled-tubing drilling brings key data in an extremely harsh environment where data is very sparse. The technology introduction in the field is a key component of a strategic vision to increase gas production through underbalance coiled-tubing drilling.

The value of this approach currently contributes to the success of UBCTD development wells and is limitless possibility of worldwide untapped reserves. Moreover, the Coil Tubing allied with this workflow can reduce carbon footprint by minimizing non-productive footage drilled and optimizing well placement investment and return of investment

Introduction

To address the challenges of formation damage caused by overbalanced drilling and enhance productivity, underbalanced coiled tubing drilling (UBCTD) is a proven solution in low mobility reservoirs (Ibrahim et al, 2024; Al-Nasser et al., 2024). However, one of the limitations of UBCTD has been the need for logging-while-drilling (LWD) technology to effectively geosteer the well within the pay zone for optimal well placement. Until recently, the only available data were LWD gamma ray (GR), cuttings analysis mineralogy analysis and correlations from offset wells.

To address this gap, a new ultra-slim LWD array resistivity tool was introduced, specifically engineered for the harsh conditions and narrow boreholes of UBCTD. This technology was integrated with a dedicated geosteering workflow that combines multi-depth resistivity measurements with offset data, machine learning models, and cuttings mineralogy to enable real-time formation interpretation. The result is a more reliable and informed approach to placing the wellbore within the most productive zones, minimizing drilling risk and maximizing reservoir exposure.

Ultraslim LWD Resistivity Technology

Operating in the challenging UBCTD environment, the ultraslim LWD resistivity (ULR) tool was designed to withstand temperatures exceeding 150 degC for 50% of the time and surpassing 125 degC for 90% of the time.

The new ULR technology (Figure 1) provides compensated electro-magnetic propagation resistivity measurements, including phase shift and attenuation at dual frequencies (2MHz and 400 kHz) and 36 in. spacing. This technology is complemented by GR, annular pressure, and annular temperature readings. With a tool outer diameter of 3 1/8 in., it operates in boreholes from 3 5/8 in. to 4 ¾ in and is rated to 350 degF.

Figure 1—Ultraslim LWD resistivity tool

The ULR tool data was validated against wireline induction data. Figure 2 presents the comparison on the left panel. The induction tool could only be conveyed with coiled tubing over a short interval at the casing exit. However, it confirms the high resistivity reading made with the ULR tool over the same interval. The other panels in Fig.2 show the resistivity and GR data in offset wells, all acquired with conventional wireline tools in 5 7/8 in. hole. The comparison between the URL and the offset wells wireline data shows a good correlation.

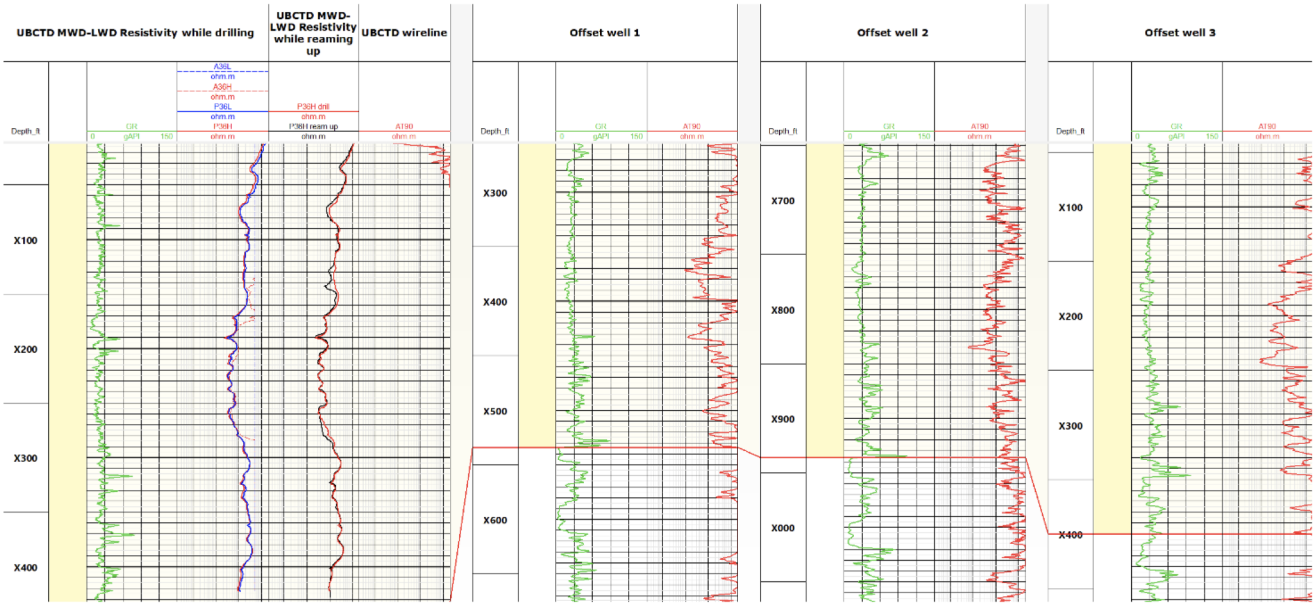

Figure 2—A good correlation is seen between the data from the ULR tool and the data from a wireline tool deployed in the same borehole (left panel), as well as the data from three offset wells within the same target formation, confirming the accuracy and reliability of the ULR measurement.

Geosteering Workflow

The resistivity measurements and GR data were combined in a model-compare-update process to steer the lateral section efficiently, enabling a thorough analysis of reservoir properties and informed decisions during the drilling operations. Figure 3 shows a schematic of the data processing workflow, which will be explained in the next sections.

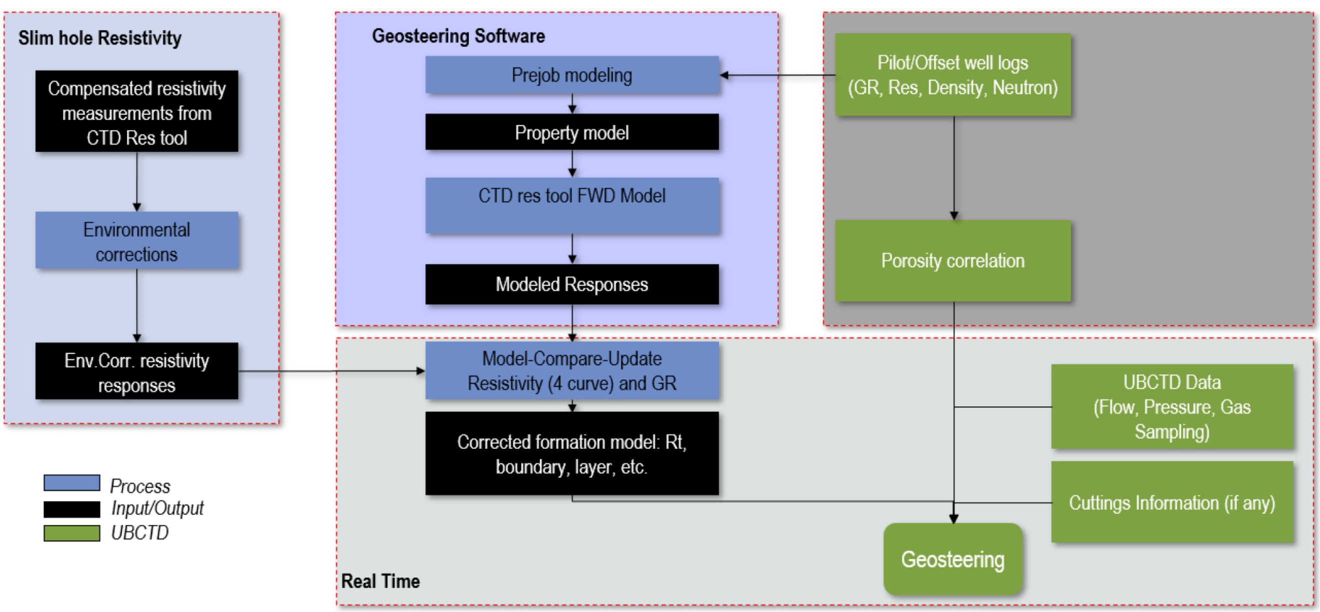

Figure 3—Overall geosteering workflow diagram. Workflow integrating ULR, pre-job modeling, and real-time model-compare-update to enable precise geosteering in UBCTD wells. Offset logs, UBCTD surface data, and cuttings information enhance formation evaluation and decision-making.

Pre Job Modeling

Before drilling commenced, a detailed pre-job modeling was performed using the resistivity curves, each associated with a different depth of investigation into the formation. The shallow resistivity is associated with the high-frequency phase shift resistivity(P36H), the medium resistivity is associated with the low-frequency phase shift resistivity (P36L), the deep resistivity is associated with the high-frequency attenuation resistivity (A36H), and the ultra-deep resistivity is associated with the low-frequency attenuation resistivity (A36L). Modeling predicts resistivity behavior across various reservoir conditions, particularly in high-resistivity and laminated zones. By simulating different drilling trajectories and formation scenarios, the expected phase shift and attenuation resistivity trends were mapped out, providing critical references for real-time steering. A key outcome was the identification of an inverse relationship between attenuation and phase shift in high-angle wells, offering valuable guidance for real-time resistivity interpretation. The workflow doesn't provide directional information, which is incorporated based on the offset wells correlation.

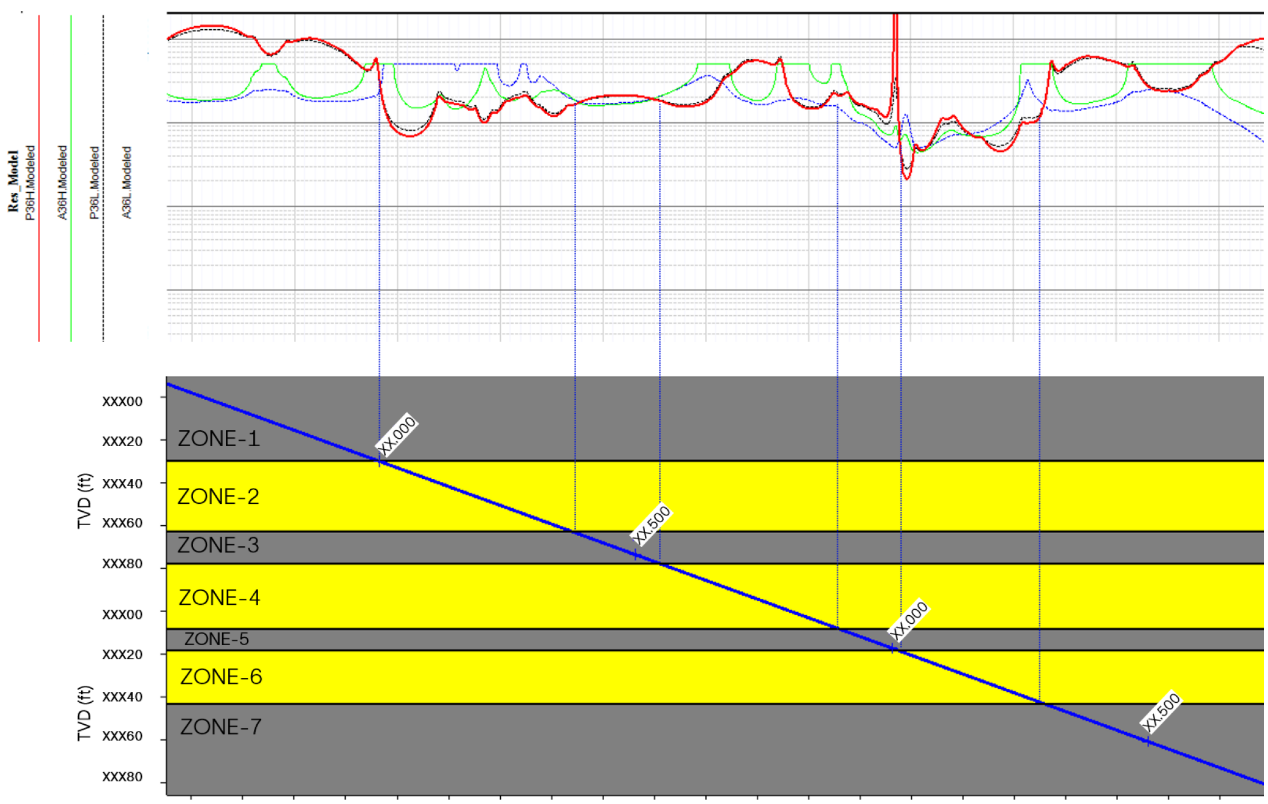

Figure 4—Pre-job modeling using resistivity curves with different depths of investigation helps identify formation layering. The curtain section was created based on 85 deg inclination, crossing several layers (Zone-1 to Zone-7). In tight layers (grey zones), the phase shift resistivity is higher than the attenuation across all frequencies. In contrast, within the target reservoir (yellow zones), the attenuation is higher than the phase shift. These modeled responses are key for supporting geosteering decisions while drilling.

The following paragraphs will discuss three geosteering model scenarios: the "base case", the "exit top" and the "exit base" scenarios.

Base Case Modeling Scenario

The base case scenario (Figure 5) shows the modeled well landing into the reservoir. It presents a simulation of how the formation might appear based on the current well plan and offset well data. The model combines petrophysical properties, formation tops, porosity indicators, and resistivity responses to support geosteering decisions. As the well approaches the reservoir, attenuation resistivity (both high and low frequencies) begins to show an increasing trend approximately 6 ft TVD before reaching the top of Zone-2. From that point, there is a vertical distance of about 15 ft to the best porosity zone. This model provides a strong reference for planning well placement and guiding real-time decisions during the landing section.

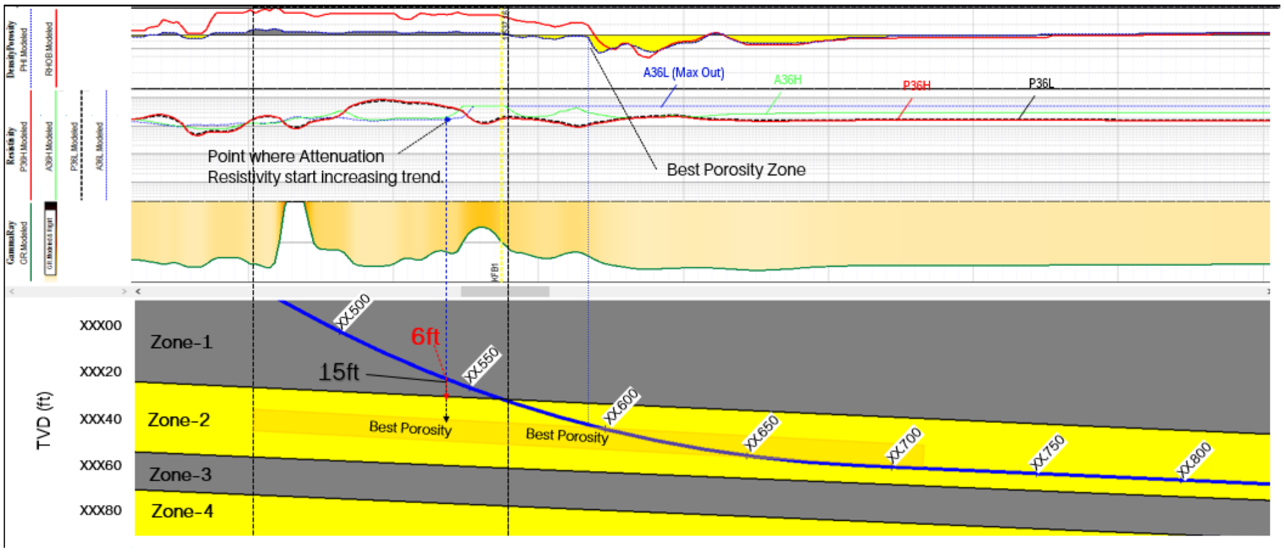

Figure 5—Base case modeling scenario - Formation tops and porosity zones are marked to guide well placement. Key markers like Zone-1, Zone-2, and Zone-4 are annotated, and the porous zones are highlighted in yellow for quick visualization of reservoir quality.

Exit Top Modeling Scenario

The "exit top" reservoir scenario (Figure 6) shows how the resistivity responses change from target reservoir to the top of reservoir. This figure illustrates a modeled scenario where the trajectory is approaching the upper boundary of the main reservoir, transitioning from Zone-2 (the primary target zone) into Zone-1 (a tighter formation). The condition shown in this model reflects a likely response as the well begins to exit upward from the reservoir.

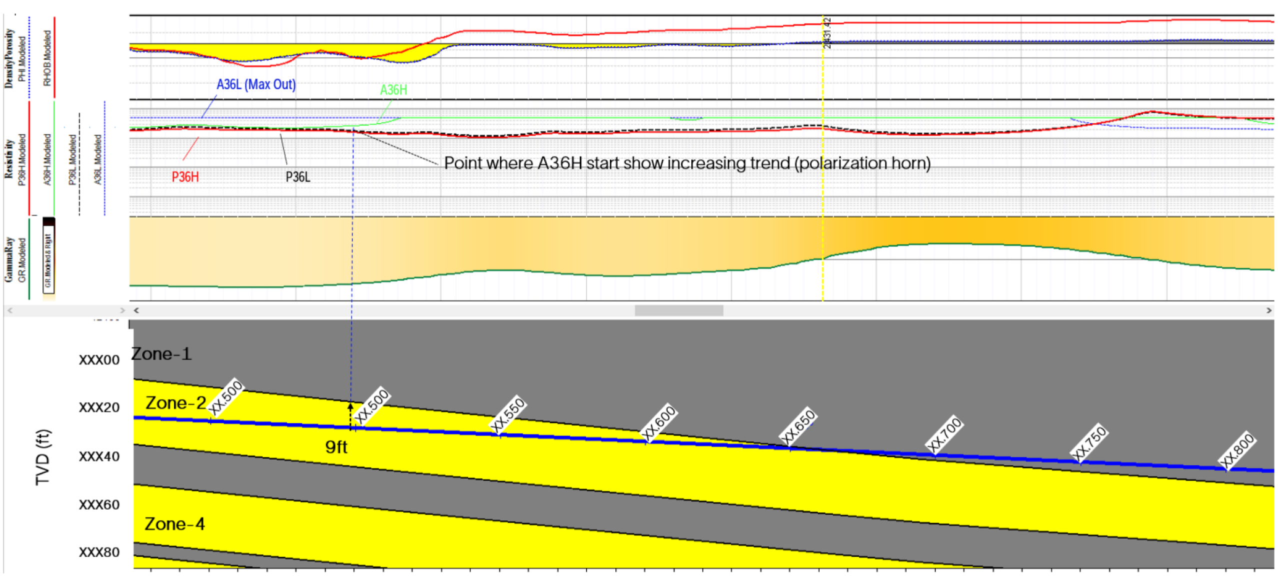

Figure 6—The Exit Top Scenarios. Each model provides a critical prediction of formation response, resistivity curve behavior, and porosity distribution to help optimize the well path during exit from the reservoir.

A key indicator is observed when the A36H attenuation curve starts showing an increasing trend, this polarization-like signature marks the point where the well is approximately 9 ft TVD below the top of Zone-2. Although it resembles a polarization horn, this specific response acts as a strong signal that the well is approaching the reservoir boundary. From a geosteering perspective, this is a critical moment. Once this resistivity behavior appears, it serves as an early warning that adjustments, such as trajectory changes, may be required to avoid exiting into the tighter formation above. Recognizing this pattern early allows for proactive geosteering decisions to maintain wellbore placement within the optimal reservoir zone.

Exit Bottom Modeling Scenario

In Figure 7, the model shows a scenario where the well trajectory exits Zone-2, our main reservoir target, and gradually enters Zone-3, a tighter formation. This transition is challenging to interpret because, while inside Zone-2, phase resistivity (both low and high frequency) consistently reads lower than attenuation, which is a typical behavior in porous zones. However, as the well approaches the tight layer of Zone-3, the phase resistivity begins to increase slowly, while attenuation remains relatively flat or even decreases slightly. Once fully within Zone-3, the phase resistivity becomes higher than attenuation, indicating entry into a less porous, tighter rock. This subtle but consistent trend reversal, where phase starts to rise above attenuation, is an important geosteering indicator. It suggests that the well is exiting a porous reservoir and moving into a tight formation. Recognizing this resistivity behavior can help guide timely trajectory corrections to avoid exiting the target zone and to maintain optimal wellbore placement.

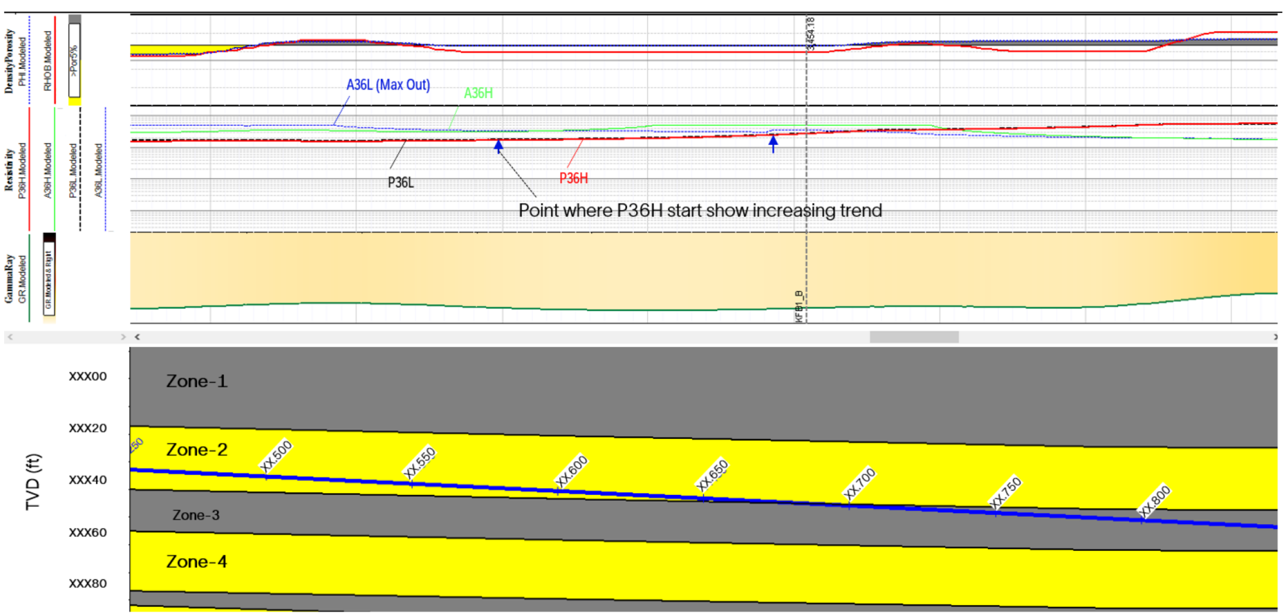

Figure 7—The Exit Bottom Scenarios. Each modeled case offers detailed guidance to anticipate formation changes, manage real-time geosteering adjustments, and optimize the final well placement as the borehole exits the bottom reservoir boundary.

Porosity Prediction Workflow

Formation porosity was estimated through a class-based machine learning model (Jain et al., 2019) using GR and resistivity data in offset wells. Wireline data (GR and deep induction resistivity) were collected from four offset wells over the formation of interest. Total porosity was computed in those wells from the formation evaluation data using a mineral solver, with the input of resistivity, bulk density, neutron porosity and neutron-induced spectroscopy data. The data was grouped into classes of similar GR and resistivity behaviors and the response was associated with the computed total porosity. The model was trained in four wells. This approach assumes that the reservoir is at irreducible water saturation conditions and that the effect of irreducible water saturation on resistivity is secondary to the effect of porosity. The process also ignores geometrical effects that could affect the electromagnetic propagation resistivity in high angle wells. This risk is mitigated using the shallowest resistivity measurement from the ULR, i.e. the high-frequency phase-shift resistivity. Based on the variability of the total porosity in each class, the process outputs a lower and upper bounds to the predicted porosity, which can be used to gauge the uncertainty on the porosity prediction.

Figure 8 shows the class-based machine learning process used to predict porosity from GR and resistivity using offset well data over the target formation. The panel on the left shows the trained model, with the input data of GR and deep induction resistivity and the output of input and predicted porosity. The panel on the right shows the GR/resistivity/porosity associations for each class. Resistivity and porosity scales are not shown but resistivity increases from left to right and porosity increases from right to left.

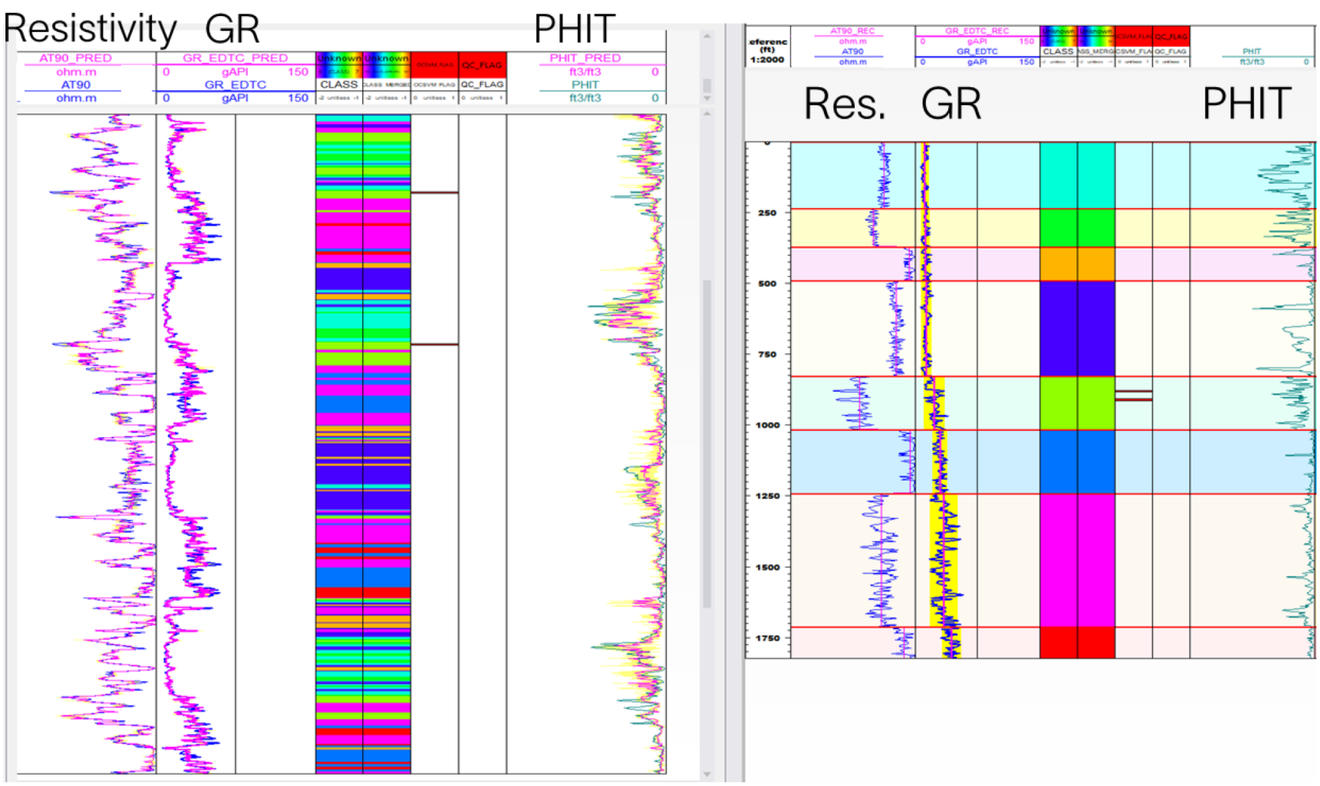

Figure 8—Class-based machine learning process used to predict porosity from GR and resistivity using offset well data over the target formation. The panel on the left shows the trained model, with the input data of GR and deep induction resistivity and the output of input and predicted porosity. The panel on the right shows the GR/resistivity/porosity associations for each class.

The trained class-based model was then applied to the data acquired in the UBCTD wells with the input of GR and ULR high-frequency phase-shift curves. The predicted porosity together with the uncertainty bounds is shown in Fig. 9 for each of the laterals.

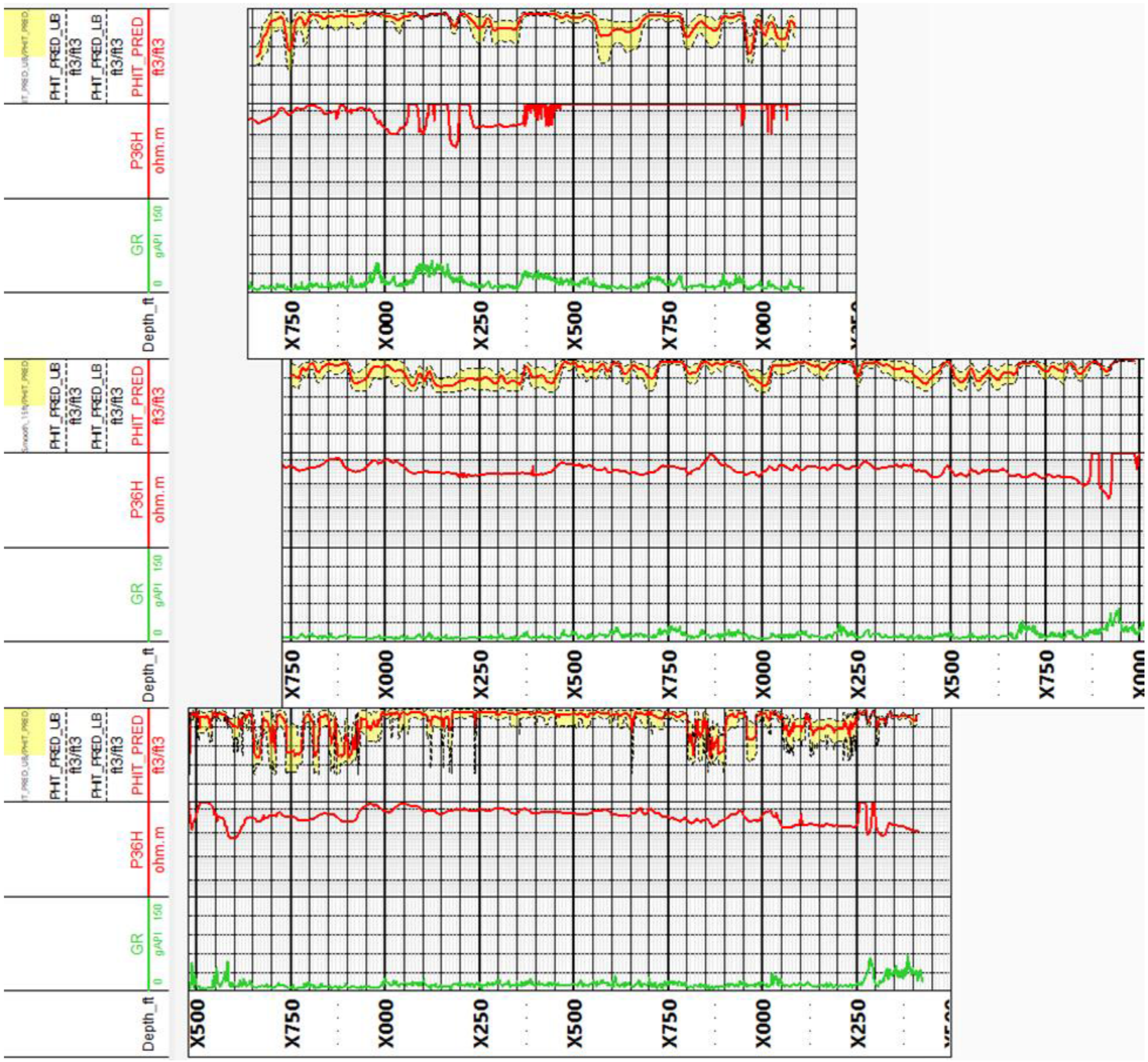

Figure 9—Predicted porosity together with the uncertainty bounds for each of the laterals. Resistivity increases from bottom to top and porosity increases from top to bottom.

Real-Time Geosteering

During drilling operations, real-time geosteering plays a critical role in navigating through these layers formation. In this project, ULR measurements were used continuously to monitor formation responses while drilling. These real-time logs were systematically compared to pre-job synthetic models, that enable validation whether the expected resistivity behavior, that was predicted during modeling, matched the actual downhole conditions. One of the key resistivity signatures observed was the correlation trend between phase and attenuation curves. The modeling phase predicted that within the reservoir zone (Zone-2), attenuation would be higher than phase, while in tighter layers (ie - Zone-3 or Zone-1), phase would increase and exceed attenuation.

This correlation held true in real-time, where the log responses aligned closely with the predicted patterns. Such alignment not only reinforced confidence in formation interpretation but also validated the model's accuracy. This continuous feedback loop between the modeled response and real-time data allowed the geosteering team to make on-the-fly adjustments with a high degree of precision. As the well approached formation boundaries or started to exit the target zone, subtle shifts in the resistivity curves, such as the emergence of a polarization horn or a reversal in phase/attenuation dominance, acted as reliable indicators. These were used to validate a well placement position and correlation in real time. Figure 10, 11 and 12 show the geosteering results in the three laterals.

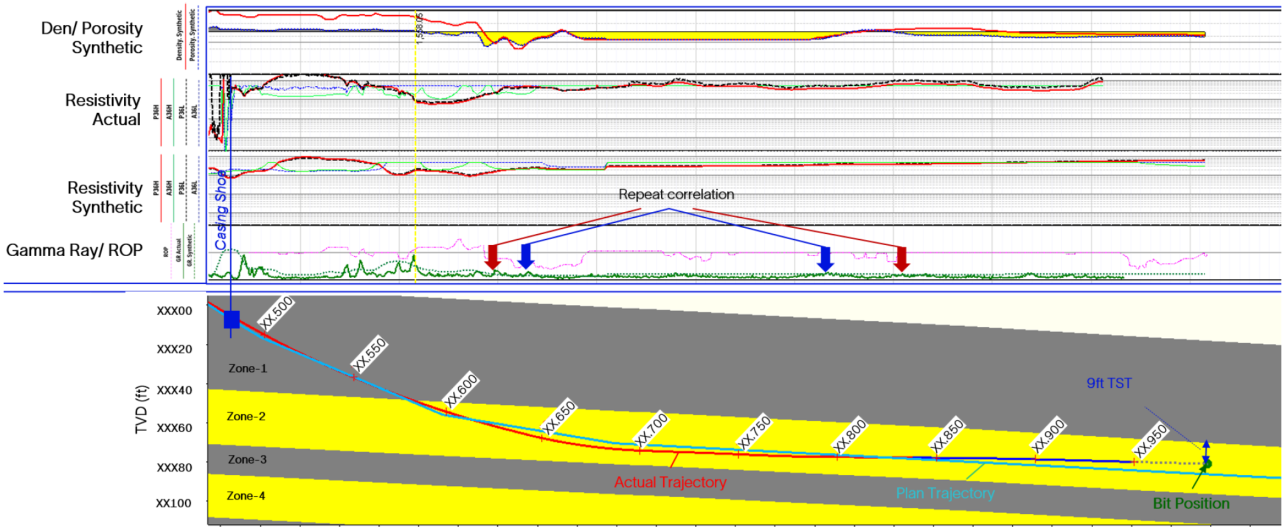

Figure 10—Application of the geosteering workflow to the drilling of the first lateral. The upper panel shows the real-time comparison of modeled synthetic vs. measured logs (GR and resistivities). Density/ Porosity was created based on synthetic log based on given placement in reservoir correlation. Repeat correlation markers confirm formation match. The lower panel shows the actual well trajectory versus plan, with a 9 ft TST (True Stratigraphic Thickness) to top of reservoir.

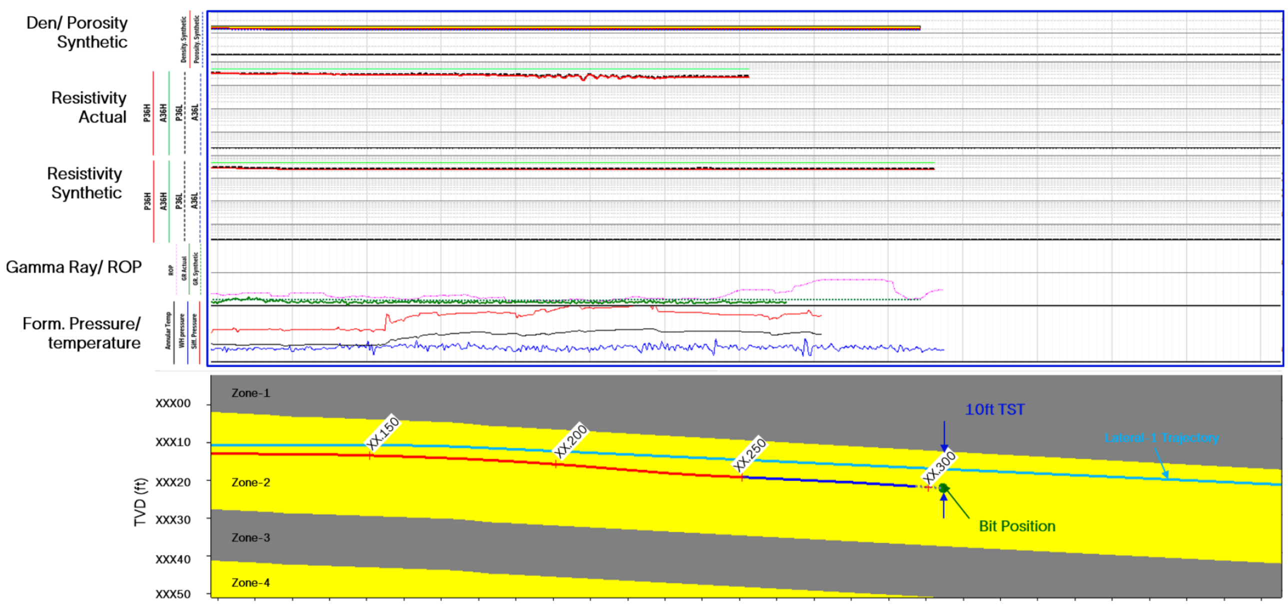

Figure 11—Second Lateral: Model-compare-update in the second lateral section shows real-time well placement validation using modeled synthetic vs. measured logs (GR and resistivities). Additional data, such as formation pressure and temperature, were integrated in this section, providing critical insight into formation behavior and gas flow response during underbalanced drilling.

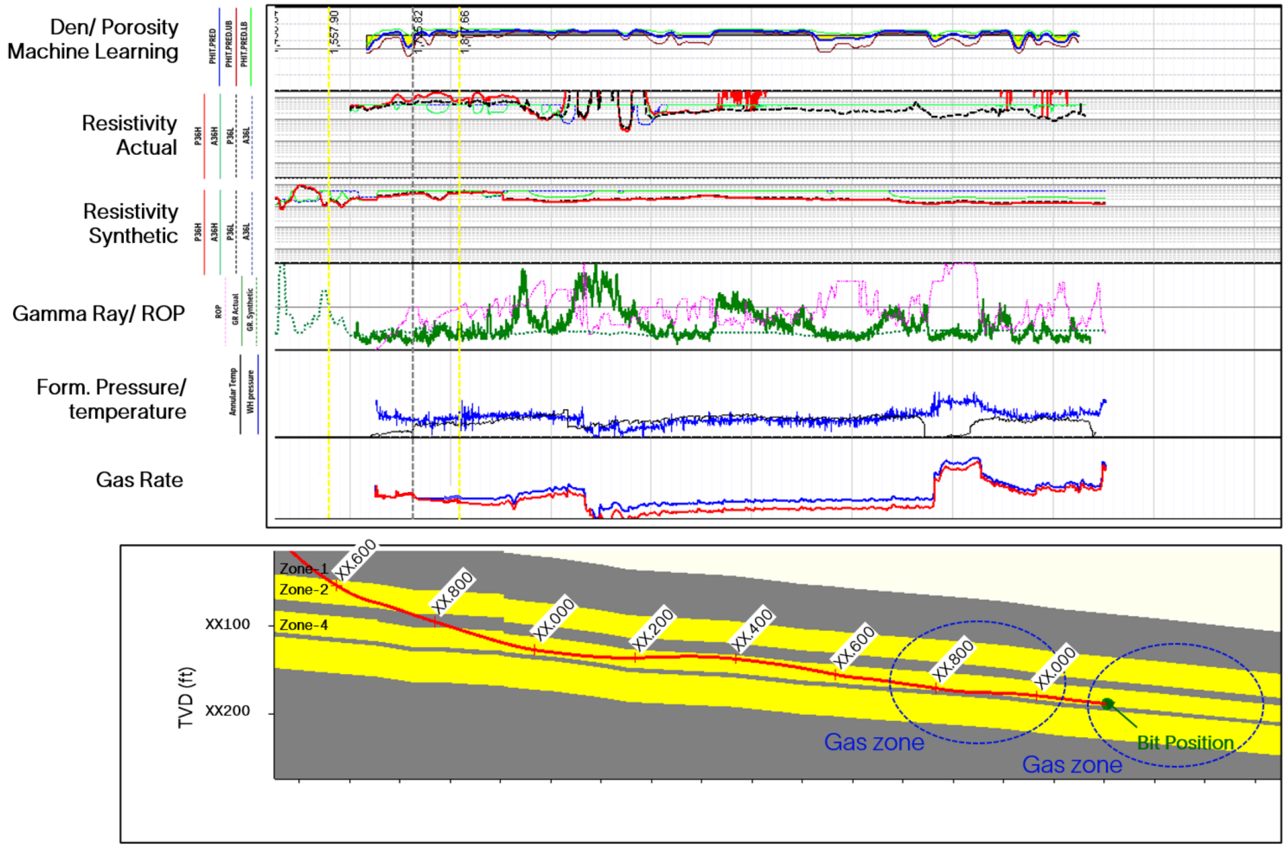

Figure 12—Third Lateral: The third lateral was steered in Zone-4. The upper track shows machine learning–predicted porosity from the real-time logs. Synthetic and actual resistivity, gamma ray, formation pressure, and temperature are integrated with gas rate measurements. Gas zones are identified along the trajectory, aiding placement decisions during underbalanced drilling.

Advanced mud logging, including mineralogy from X-ray fluorescence (XRF) and diffraction (XRD), and production and drilling mechanics indicators further strengthened the interpretation of subsurface variability. These real-time measurements provided added confidence to correlation of reservoir boundaries, especially when signal contrasts were subtle, or zone transitions occurred rapidly.

The addition of mineralogy data enhances the structural correlation process and provides confidence on the geosteering results. Figure 13 shows a comparison between offset well neutron-induced spectroscopy mineralogy and deep resistivity, with mud-logging XRD and ULR resistivity in two UBCTD laterals. The offset well is a sub-vertical well, while the UBCTD wells are high-angle wells. In both laterals, the initial part of the well is below 90 deg. inclination and hence crosses the formation sequence top-down. The change from a highly dolomitic rock containing a fraction of anhydrite is well correlated between the offset well and the two UBCTD wells.

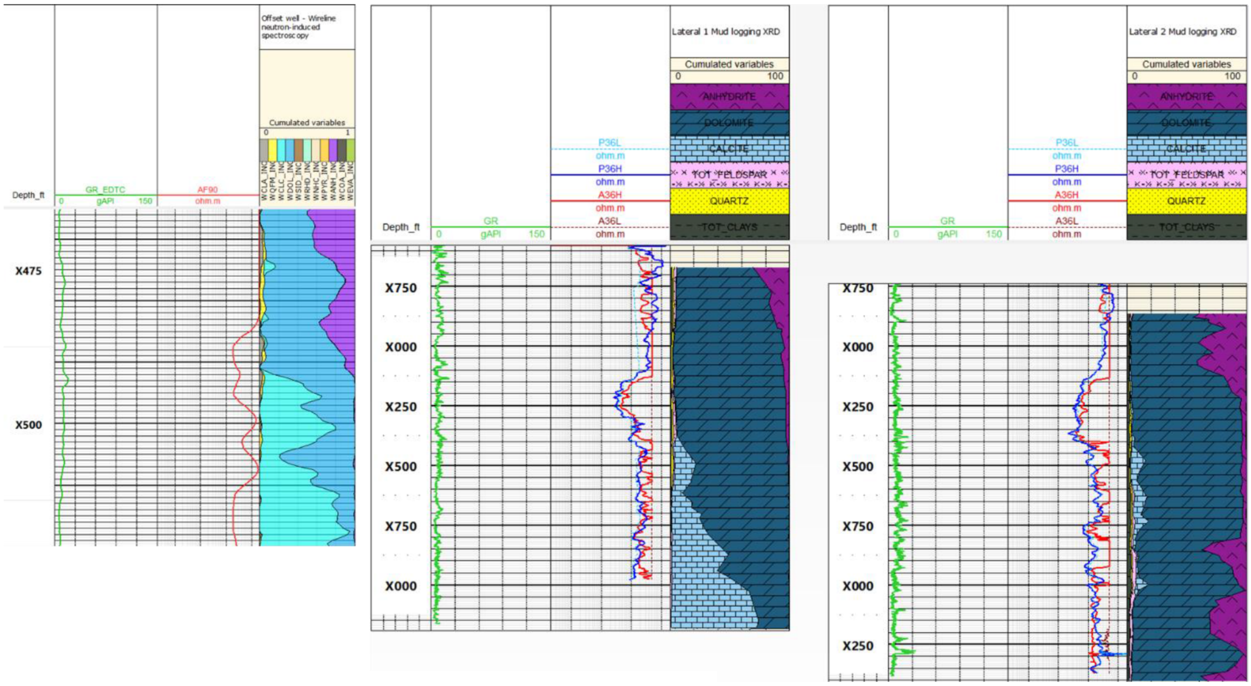

Figure 13—Comparison of offset well neutron-induced spectroscopy mineralogy and deep resistivity (left) with mud-logging XRD and ULR resistivity in two UBCTD laterals (right).

Conclusions

Advances in LWD logging for UBCTD applications with the introduction of the ULR tool has enabled the development of a geosteering workflow to optimize the placement of wells drilled with UBCTD ultra-slim holes. By gaining deeper insights into geological variability, supported by data integration with mud logging, production data, offset wells data and drilling parameters, machine learning, and real-time feedback loops, the geosteering team was able to achieve excellent wellbore placement across all laterals. This approach ensured improved reservoir contact, contributing to maximum hydrocarbon recovery and overall operational efficiency.

References

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